

Selected Machine Learning Labs

Graph Learning · Optimization for Machine Learning · Machine Learning

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1. Graph Learning

A guided laboratory sequence connecting classical graph algorithms with representation learning and graph neural networks. This summary uses only notebooks with executed, error-free outputs.

89.88%

LABEL DIFFUSION · TEXT

86.90%

LABEL DIFFUSION · LINKS

82.95%

GNN · WIKIVITALS TEST

- Implemented and analyzed PageRank and personalized PageRank on directed and bipartite graphs.
- Studied Louvain community detection, modularity, Adjusted Rand Index, and the Simpson index.
- Compared text-based, link-based, and combined label diffusion; explored spectral embeddings.
- On WikiVitals, the GNN reached 82.95% test accuracy, compared with 81.13% for logistic regression and 74.50% for heat diffusion.

PageRank

Louvain

Diffusion

Spectral Embedding

GNN

Python

2. Optimization for Machine Learning

Stochastic gradient descent on MNIST

An individual laboratory implementing and comparing a custom stochastic-gradient routine with Keras across learning rates, batch sizes, and training schedules.

95.39%

BEST RECORDED TEST
ACCURACY

26,506

MODEL PARAMETERS

5

EPOCHS · BEST RUN

- Compared optimization behaviour across batch size and learning-rate configurations.
- The strongest stored run used batch size 10 and learning rate 1 for five epochs.

SGD

Keras

MNIST

Python

3. Machine Learning

SVMs: kernels, overfitting, and PCA

An individual lab comparing linear and RBF support-vector machines, regularization, and learning curves. On the LFW face task, PCA preserved 93.4% accuracy while reducing fit time from 307.7 seconds to 2.48 seconds (approximately 124×).

MNIST: linear classifier and autoencoders

A NumPy softmax classifier achieved 88.42% held-out test accuracy, followed by experiments with PCA, autoencoders, and denoising autoencoders in Keras.

Scikit-learn

SVM

PCA

NumPy

Keras

Autoencoders

Scope and quality note. This document is a sanitized portfolio summary, not a replacement for the original coursework. Raw notebooks and datasets are intentionally not redistributed. Incomplete or error-producing exercises and unverified metrics were excluded.